

Aditya Dasika

Machine Learning Engineer · Multi-Agent Orchestration on LangGraph · Conversational AI & Graph RAG · LLM Evaluation & Serving

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EXPERIENCE

AI/LLM Engineer II

April 2025 – Present

Phenom People

Hyderabad, Telangana

- Architected **chatbotCXA**gent, a production multi-agent conversational system on **LangGraph** whose orchestration layer coordinates 12 nodes spanning intent recognition, planning, dialogue-state management and tool routing, with **Graph RAG** information retrieval over enterprise knowledge and document repositories via FAISS vector search and a Neo4j graph for multi-hop, policy-aware grounding.
- Engineered tool-calling agents that execute governed actions — database queries and structured extraction from unstructured documents — under **XGrammar constrained decoding** and schema-validated JSON guardrails, raising tool-routing accuracy to **91.9%** and eliminating malformed tool calls in live traffic.
- Hardened the fine-tuned Qwen deployment for production serving — required tool-choice and temperature pinned at 0.1 for deterministic tool calls — and held it to a golden baseline of 1.0s median and **1.4s** P95 latency with zero failures across **5,510** requests, taking the service's test coverage from **7% to 78%**.
- Built an **LLM evaluation framework** (pass@k, LLM-as-judge rubric) gating every release across all 12 nodes, instrumented with LiteLLM and Langfuse for per-node cost, latency and full-trace auditability — continuously monitoring agent behaviour and curbing hallucinations.
- Fine-tuned and served Qwen models via **QLoRA** (Unsloth, NF4) on **vLLM** / AWS g5.xlarge (A10G), sustaining 10.98 RPS at 98.6% success under load at roughly half the inference cost of hosted GPT-4.1.
- Fine-tuned a multi-task **ModernBERT** (intent + NER) in PyTorch, raising NLU accuracy from 87% to 97%.
- Developed production-grade Python services — conversation analytics APIs covering both live-agent and WhatsApp channels — raising test coverage from **63% to over 90%** and clearing static-analysis code smells for clean, maintainable code, and migrated the service to **FastAPI**.

Full Stack Engineer

May 2022 – February 2025

Blueleaves Farms

Hyderabad, Telangana

- Owned end-to-end delivery of production systems (POS, CRM, inventory, procurement, e-commerce) on Node.js, Firebase Cloud Functions and REST APIs — with role-based access control and CI/CD — integrating multiple internal systems and cutting manual operational errors by roughly 90%.
- Built data-driven analytics on large-scale transactional data — demand forecasting and inventory modeling across 5+ locations and 10,000+ monthly transactions.

PROJECTS

IndexNotes — AI Knowledge Graph for Scientific Papers | *Neo4j, Ollama, PyMuPDF, Python, Docker*

- Built an automated Graph RAG pipeline ingesting research papers (PDF to local-LLM extraction to structured JSON to Neo4j), creating 40+ concepts and 17+ typed relationships, with quantized local models via Ollama for entity and relationship extraction.
- Designed a Neo4j schema enabling multi-hop graph traversal and cross-document information retrieval across scientific domains.

NorthStar — Motorcycle Navigation Companion App | *Android, BLE, Maps SDK*

- Built and shipped a companion app for the **Royal Enfield** Tripper Dash, delivering turn-by-turn navigation to the bike's connected instrument cluster and growing an organic user base without paid distribution; the traction drew **Royal Enfield's** attention directly, leading to an on-site visit at their HQ and a continuing relationship.

TECHNICAL SKILLS

Generative AI, LLMs & Agentic Tools: Agentic / Multi-Agent Systems, LangGraph, LangChain, Tool/Function Calling, Prompt Engineering, Model Routing, Context-Window Management, RAG & Graph RAG, LLM Fine-Tuning (QLoRA, SFT)

Languages: Python, JavaScript, TypeScript, SQL, Bash

Eval, Observability & Guardrails: LLM Evaluation (LLM-as-judge, pass@k), OpenTelemetry, Constrained Decoding (XGrammar), Schema Validation, Langfuse, LiteLLM, Model Monitoring, Hallucination Mitigation, Role-Based Access Control

ML, Deep Learning & Serving: PyTorch, Transformers (BERT/ModernBERT), HuggingFace PEFT, Unsloth, vLLM, Embeddings / Vector Search (FAISS), Neo4j, Ollama, Production MLOps

Backend & Infra: FastAPI, Node.js, REST APIs, Microservices, Docker, Helm, AWS (EC2/S3/SageMaker), GitHub Actions, CI/CD, Linux

Databases: PostgreSQL, Neo4j, MongoDB, ClickHouse, Redis, Firebase

EDUCATION

Andhra Loyola Institute of Engineering and Technology

Bachelor of Technology in Mechanical Engineering

Vijayawada, Andhra Pradesh

August 2018 – May 2022